

Assignment IV – Sparse S&P 500 Portfolio Replication

DA6701 — Benchmark: ^GSPC daily log returns — Train: Jan 2020–Dec 2024 — Validation: Jan 2025–Jun 2025 — Holdout: Jul 2025–Dec 2025

578 Readable CSVs	553 Final Stocks	31 Excluded Files / Names	GA Best Method	4.63% Holdout TE @ k=50	-0.10 Holdout IR @ k=50
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1 — Problem Setup, Data Pipeline, and Model Design

The task is to replicate the S&P 500 daily return profile with a sparse long-only portfolio under the cardinality constraint $k \leq 50$. After validating the raw OHLCV files, we retained 578 readable stock histories, excluded 6 issue files (5 empty CSVs and one non-price cache file), removed 24 late entrants / early exits, and dropped one additional ticker (CVG) because it still contained 25 missing training prices. The final model-ready matrix covers 2020-01-03 to 2025-12-31 and contains 553 stocks plus the benchmark. Validation and holdout price holes were forward-filled before return construction, while the training block was kept hole-free by design.

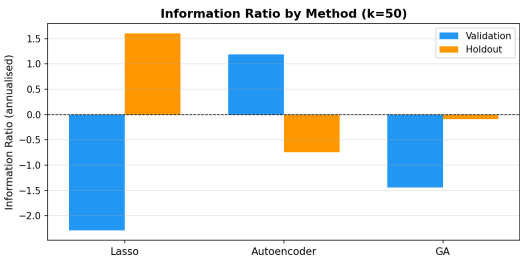
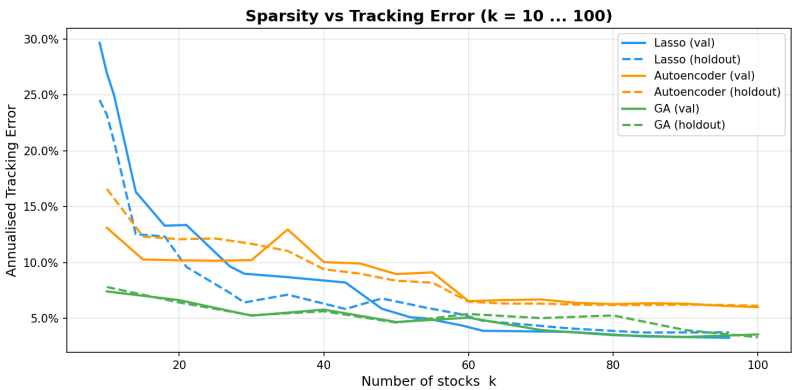
The cardinality-constrained tracking problem is an MIQP and is therefore NP-hard. To keep the comparison fair, all methods use the same two-stage pipeline: (i) method-specific stock selection and (ii) a shared long-only quadratic program that minimizes the variance of active returns subject to $\sum_i w_i = 1$ and $w_i \geq 0$. We report annualized Tracking Error (TE) and Information Ratio (IR), where TE should be low and IR should stay close to zero for a true index tracker.

Method	Selection Stage	Implementation Note
Lasso	Non-negative Lasso regression on training benchmark returns; alpha swept on a log-grid to produce sparse portfolios of different k .	Fast, interpretable, and directly supervised by the benchmark, but sensitive to regime shocks and concentration.
Autoencoder	Symmetric autoencoder ($553 \rightarrow 128 \rightarrow 32 \rightarrow 128 \rightarrow 553$); stocks ranked by communality (reconstruction R^2) and top- k retained.	Captures shared latent structure, but unsupervised ranking can over-concentrate in sectors with strong common factor loadings.
Genetic Algorithm	Population search over k -stock subsets; fitness equals negative training TE, with crossover, mutation, and shared QP reweighting.	Computationally heavier, but best at directly optimizing sparse subset quality under the replication objective.

2 — k = 50 Summary Results

Method	TE Val	IR Val	TE Holdout	IR Holdout	Interpretation
Lasso	5.86%	-2.29	6.78%	+1.60	Moderate tracking error, but a large IR sign flip reveals unstable active tilts under the April 2025 tariff shock and subsequent rebound.
Autoencoder	8.96%	+1.19	8.37%	-0.74	Highest TE at $k = 50$; the unsupervised ranking produces persistent sector concentration, so the portfolio behaves less like a neutral tracker.
GA	4.68%	-1.44	4.63%	-0.10	Best overall. It delivers the lowest out-of-sample TE on both splits and the holdout IR closest to the ideal target of zero.

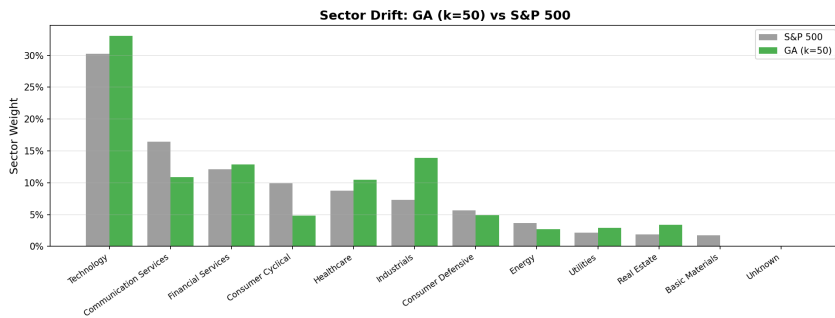
3 — Required Deliverables: Sparsity vs Tracking Error and Information Ratio



Out-of-sample TE falls steeply as k rises from 10 to roughly 50, then flattens. GA dominates the sparse end of the trade-off curve. IR quantifies active drift. For a tracker, values near 0 are preferred; GA is closest on holdout.

The deliverable curve shows the expected diminishing-return pattern: all methods improve rapidly from $k = 10$ to around $k = 50$, after which TE reductions become incremental. Lasso is especially brittle at small k (TE above 25% at $k \approx 10$), while GA already remains below 8% at $k = 10$, indicating better subset quality under severe sparsity. The IR plot adds the second required deliverable by highlighting that low TE alone is not sufficient: Lasso and Autoencoder both exhibit unstable directional bets across validation and holdout, whereas GA's holdout IR of -0.10 is the clearest evidence of genuine replication rather than accidental active management.

4 — Required Deliverable: Sector Drift at $k = 50$



The required sector-drift chart is shown for the GA portfolio at $k = 50$, because this is the best overall tracker. Its sector weights remain close to the market-cap weighted S&P 500 baseline, with only a mild Technology overweight. This makes the portfolio much more defensible as a replication strategy than the alternatives.

Lasso partially satisfies the drift criterion but still overweights Healthcare and underweights Communication Services. Autoencoder fails the sector-neutrality test: it concentrates heavily in Financial Services and Utilities because communality ranking favors stocks explained by the same latent factors.

5 — Conclusion

Across all required deliverables, the **Genetic Algorithm** is the strongest replication method. It achieves the lowest validation and holdout TE at the assignment target of $k = 50$, keeps holdout IR closest to zero, and exhibits the cleanest sector alignment with the benchmark. Lasso remains attractive for speed and interpretability but is vulnerable to regime-specific tilts, while the Autoencoder's unsupervised selection objective produces excessive sector concentration. The main empirical takeaway is that, for sparse index replication, directly optimizing subset quality with a search-based method and then reweighting through a common long-only QP yields the most stable out-of-sample tracker.